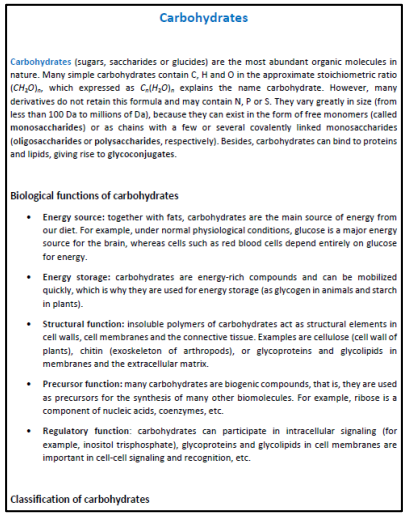


Carbohydrates

Structure and function

Supportive material



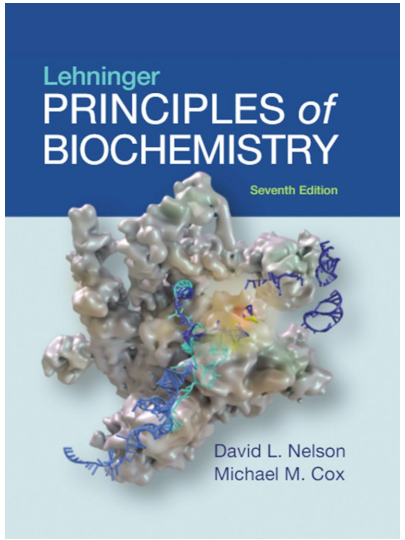
Carbohydrates

Carbohydrates (sugars, saccharides or glucides) are the most abundant organic molecules in nature. Many simple carbohydrates contain C, H and O in the approximate stoichiometric ratio $(CH_2O)_n$, which expressed as $C_n(H_2O)_n$ explains the name carbohydrate. However, many derivatives do not retain this formula and may contain N, P or S. They vary greatly in size (from less than 100 Da to millions of Da), because they can exist in the form of free monomers (called **monosaccharides**) or as chains with a few or several covalently linked monosaccharides (**oligosaccharides or polysaccharides**, respectively). Besides, carbohydrates can bind to proteins and lipids, giving rise to **glycoconjugates**.

Biological functions of carbohydrates

- **Energy source:** together with fats, carbohydrates are the main source of energy from our diet. For example, under normal physiological conditions, glucose is a major energy source for the brain, whereas cells such as red blood cells depend entirely on glucose for energy.
- **Energy storage:** carbohydrates are energy-rich compounds and can be mobilized quickly, which is why they are used for energy storage (as glycogen in animals and starch in plants).
- **Structural function:** insoluble polymers of carbohydrates act as structural elements in cell walls, cell membranes and the connective tissue. Examples are cellulose (cell wall of plants), chitin (exoskeleton of arthropods), or glycoproteins and glycolipids in membranes and the extracellular matrix.
- **Precursor function:** many carbohydrates are biogenic compounds, that is, they are used as precursors for the synthesis of many other biomolecules. For example, ribose is a component of nucleic acids, coenzymes, etc.
- **Regulatory function**: carbohydrates can participate in intracellular signaling (for example, inositol trisphosphate), glycoproteins and glycolipids in cell membranes are important in cell-cell signaling and recognition, etc.

Classification of carbohydrates



Additional study material

Aula Virtual:

- Study material
- Video
- Exercises

Reading

Lehninger’s PoB Chapter 7

Learning goals

- General characteristics and biological functions of carbohydrates
- Structure and names of some representative carbohydrates
- How to identify monosaccharides in the Haworth projection
- Stereoisomerism in monosaccharides
- How glucose is measured in blood
- The O-glycosidic bond
- The biological relevance of α1 → 4 vs. β1 → 4 glycosidic bonds

Structure determines function

General characteristics

- Most abundant biomolecules in nature
- Named so because many have the formula $C_n(H_2O)_n$
- Also known as sugars, glucides or saccharides
- Range from glyceraldehyde (M_w = 90 g/mol) to amylopectin (M_w > 200,000,000 g/mol)
- Can be covalently linked to proteins or lipids

Biological functions

- Energy source and storage
- Structural components
- Precursors for other biomolecules
- Regulators in cell-cell signaling and recognition

Classification

- Monosaccharides

  Single sugar units

- Oligosaccharides (disaccharides, glycoconjugates)

  Short chains of few monosaccharides

- Polysaccharides

  Polymers of many monosaccharides

Monosaccharides: nomenclature

- Commun names usually end in *-ose*
- Based on the number of carbon atoms

  Trioses, tetroses, pentoses, hexoses…

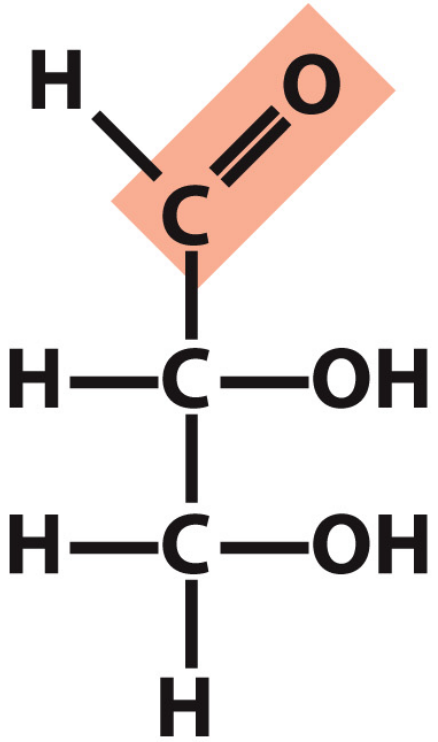
- Based on the functional group

  Aldoses and ketoses

Monosaccharides: basic structure

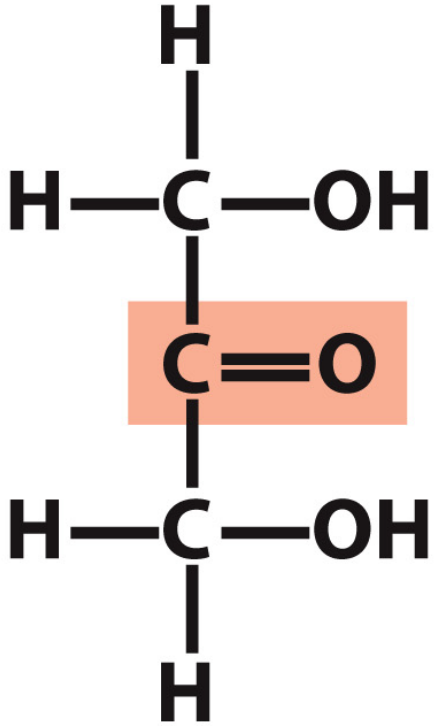
ALDOSES

(polyhydroxy aldehyde)



KETOSES

(polyhydroxy ketone)



Monosaccharides: isomerism

- Functional isomerism
  - Different functional group (aldehyde or ketone)
- Stereoisomerism
  - Different 3D orientation of atoms
    - Enantiomers
      - Non-superimposable mirror images
    - Diastereomers
      - Non-mirror-image stereoisomers
    - Epimers
      - Different orientation of only one asymmetric carbon
    - Anomers
      - Different orientation of the anomeric carbon

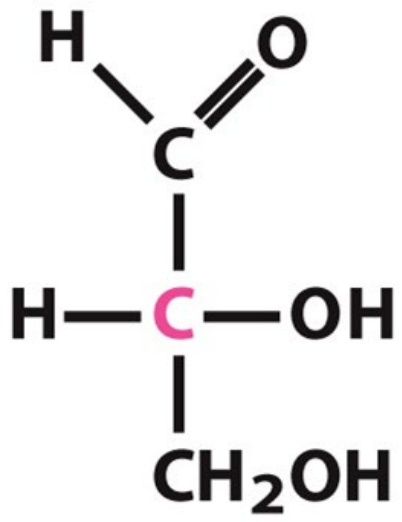


Watch this video about isomerism in monosaccharides

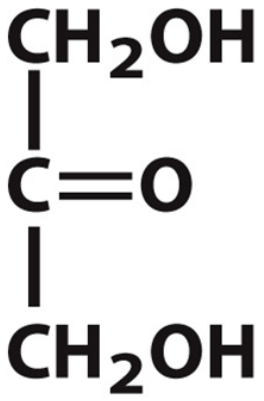
Monosaccharides: classification

	Aldose	Ketose
C3	Aldotriose	Ketotriose
C4	Aldotetrose	Ketotetrose
C5	Aldopentose	Ketopentose
C6	Aldohexose	Ketohexose
C7	Aldoheptose	Ketoheptose

Monosaccharides: trioses



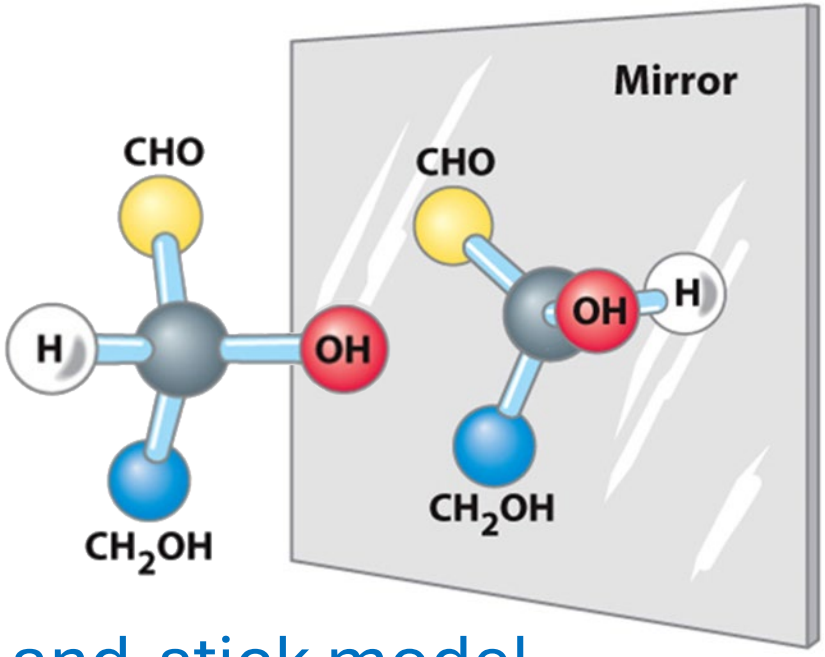
Glyceraldehyde



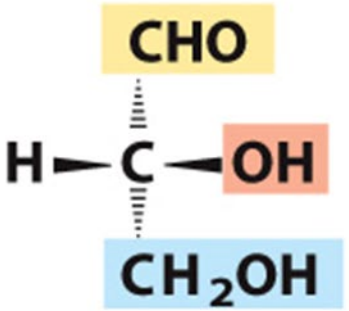
Dihydroxyacetone

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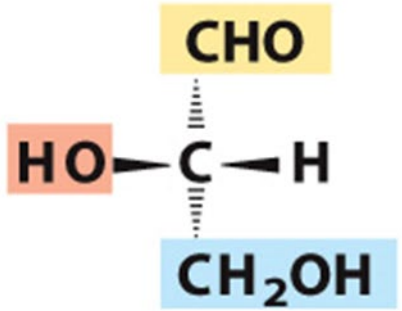
Monosaccharides: D- and L-glyceraldehyde



Ball-and-stick model

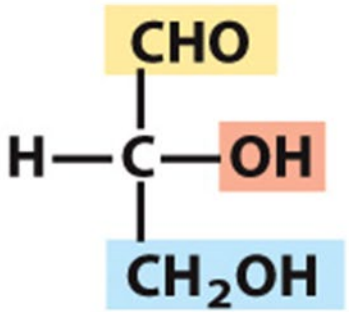


D-Glyceraldehyde

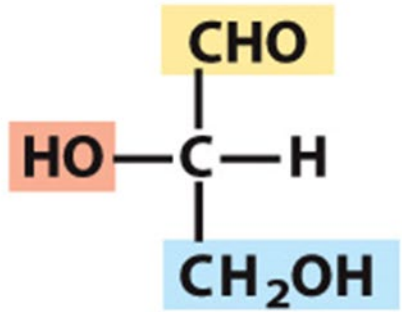


L-Glyceraldehyde

Fischer projection



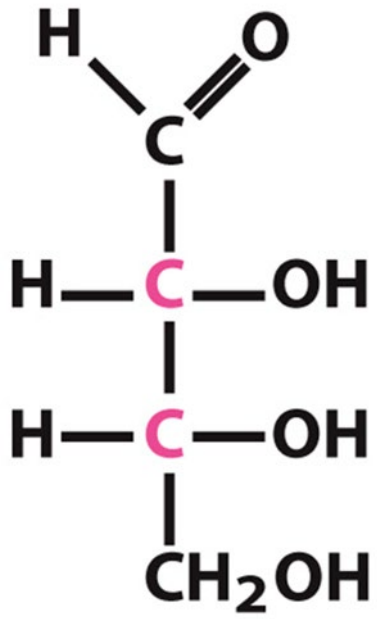
D-Glyceraldehyde



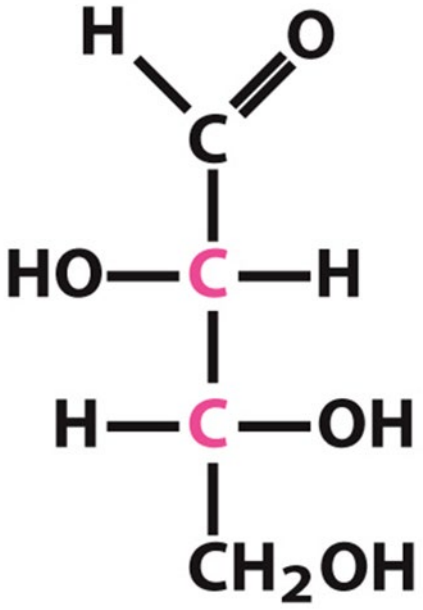
L-Glyceraldehyde

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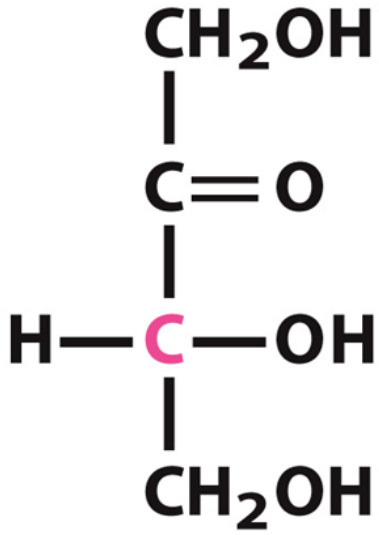
Monosaccharides: Tetroses



D-Erythrose



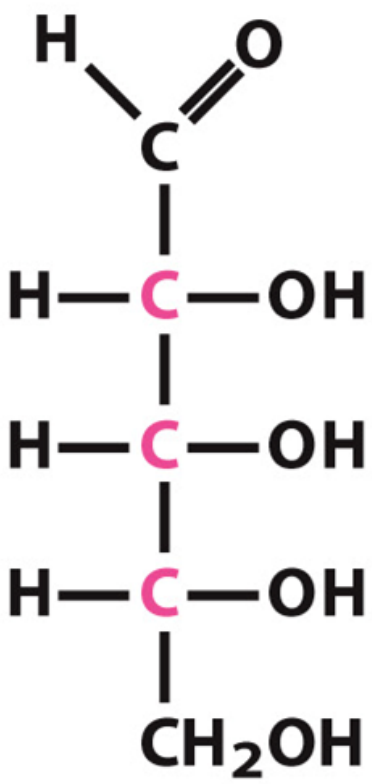
D-Threose



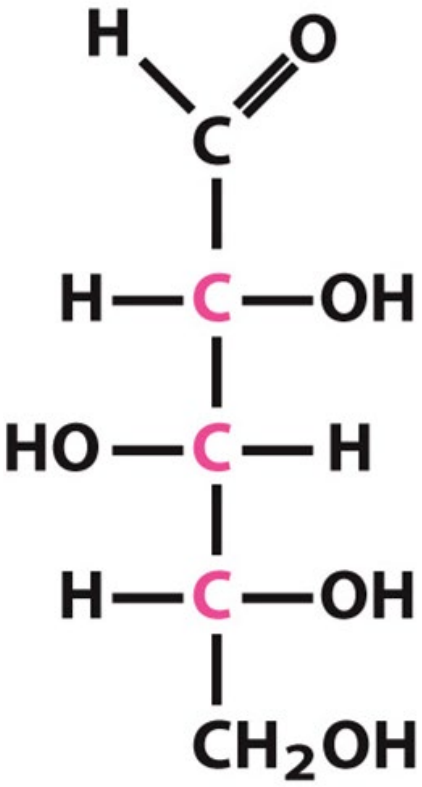
D-Erythrulose

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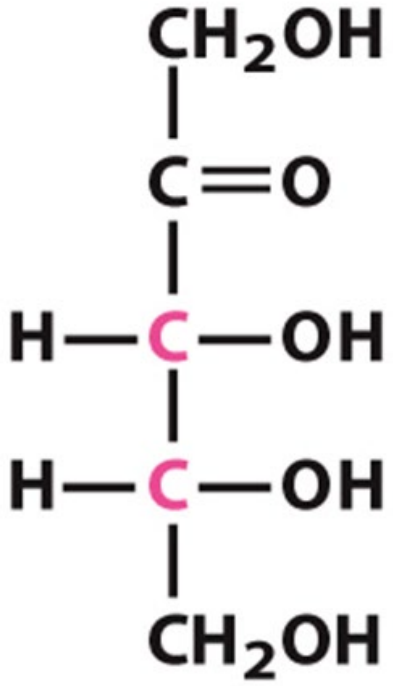
Monosaccharides: Pentoses



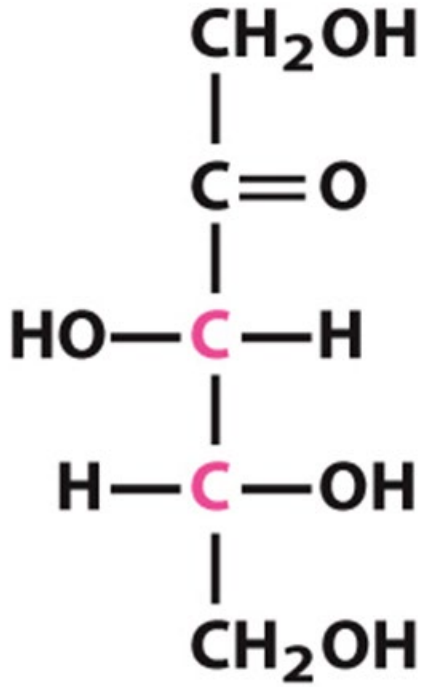
D-Ribose



D-Xylose



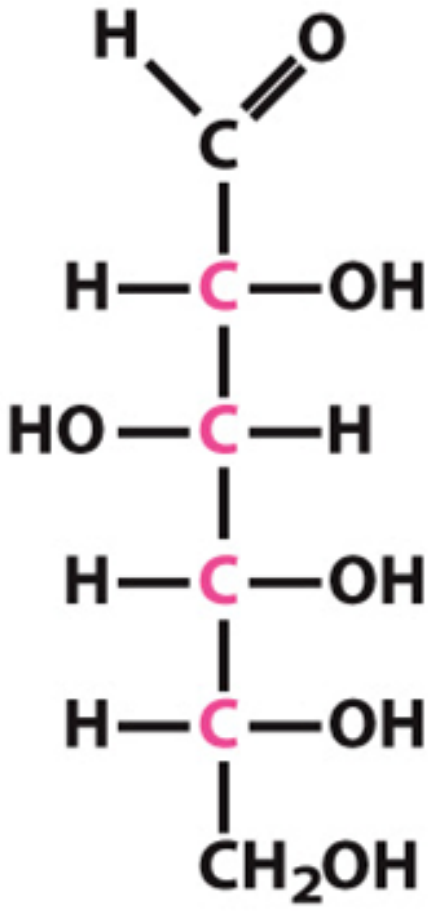
D-Ribulose



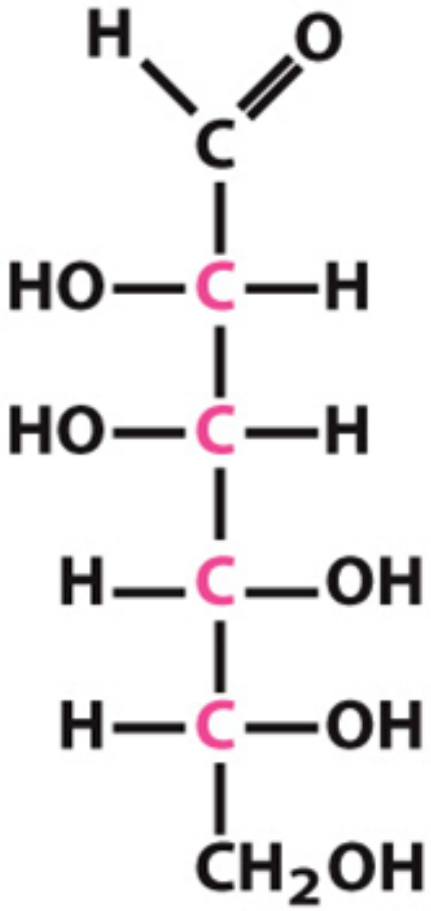
D-Xylulose

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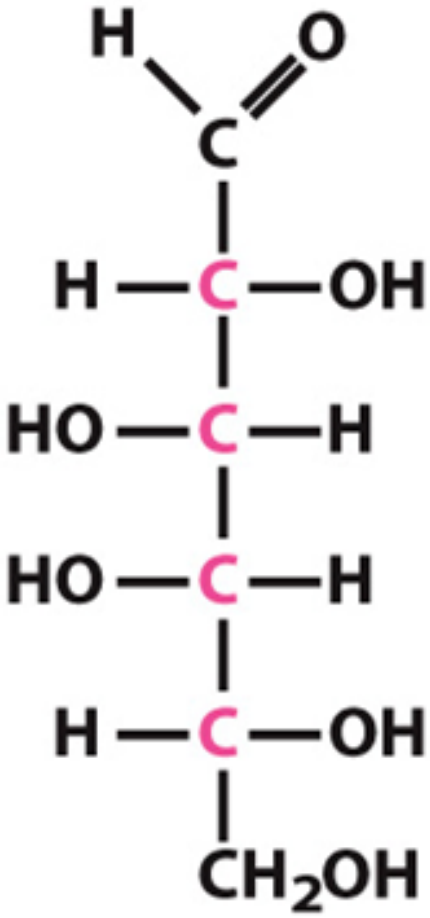
Monosaccharides: Hexoses



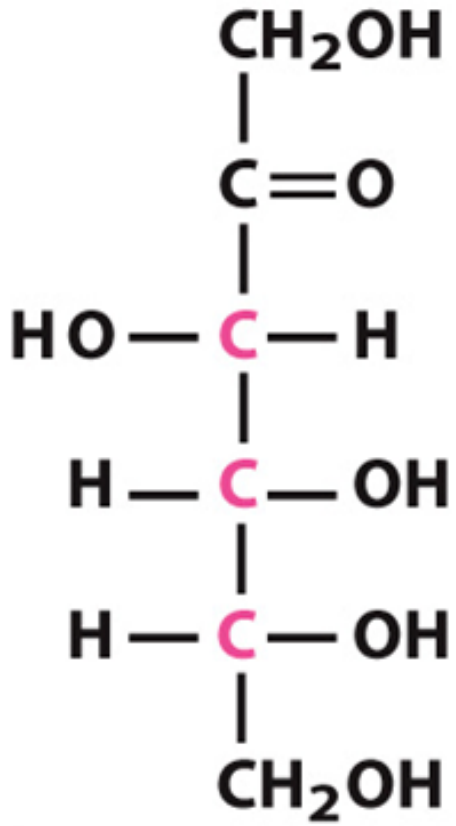
D-Glucose



D-Mannose



D-Galactose

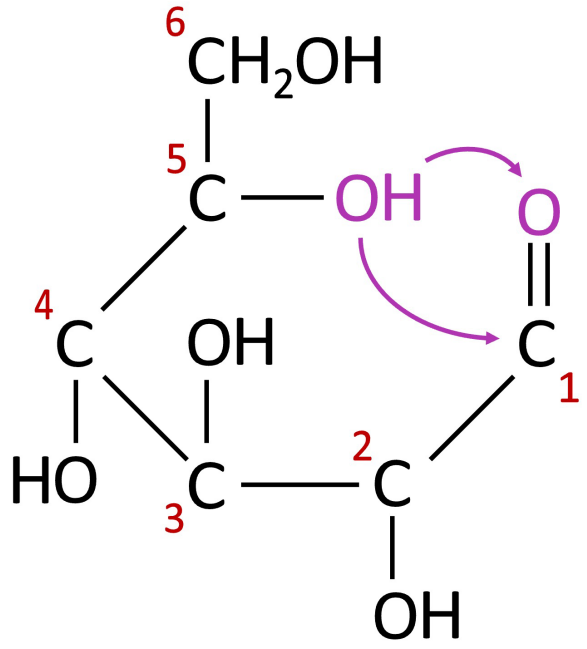


D-Fructose

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Monosaccharide cyclization

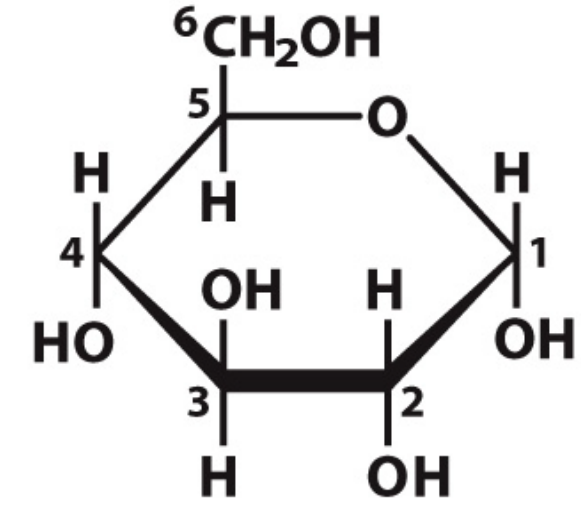
Although we frequently represent monosaccharides as straight, open-chain structures, monosaccharides with five or more carbon atoms exist preferentially as ring forms in aqueous solution



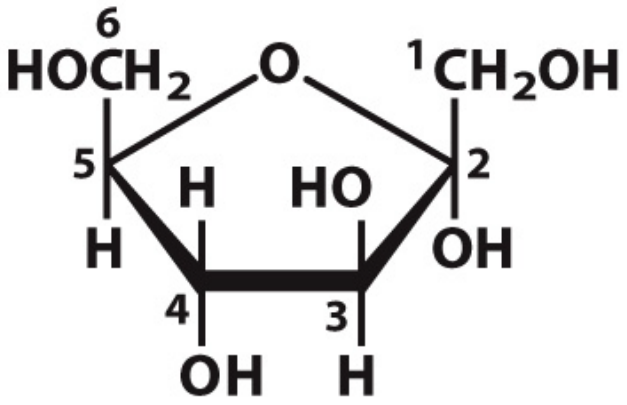
Monosaccharide cyclization

Six-membered oxygen-containing rings are called pyranoses

Five-membered oxygen-containing rings are called furanoses



α-D-Glucopyranose



α-D-Fructofuranose

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Monosaccharide cyclization

- The carbonyl carbon reacts with a hydroxyl group and becomes a new chiral center known as the anomeric carbon
- Two anomers are possible, depending on the orientation of the hydroxyl group at the anomeric carbon

  α anomer: on the opposite side of the $-CH_2OH$ group

  β anomer: on the same side as the $-CH_2OH$ group

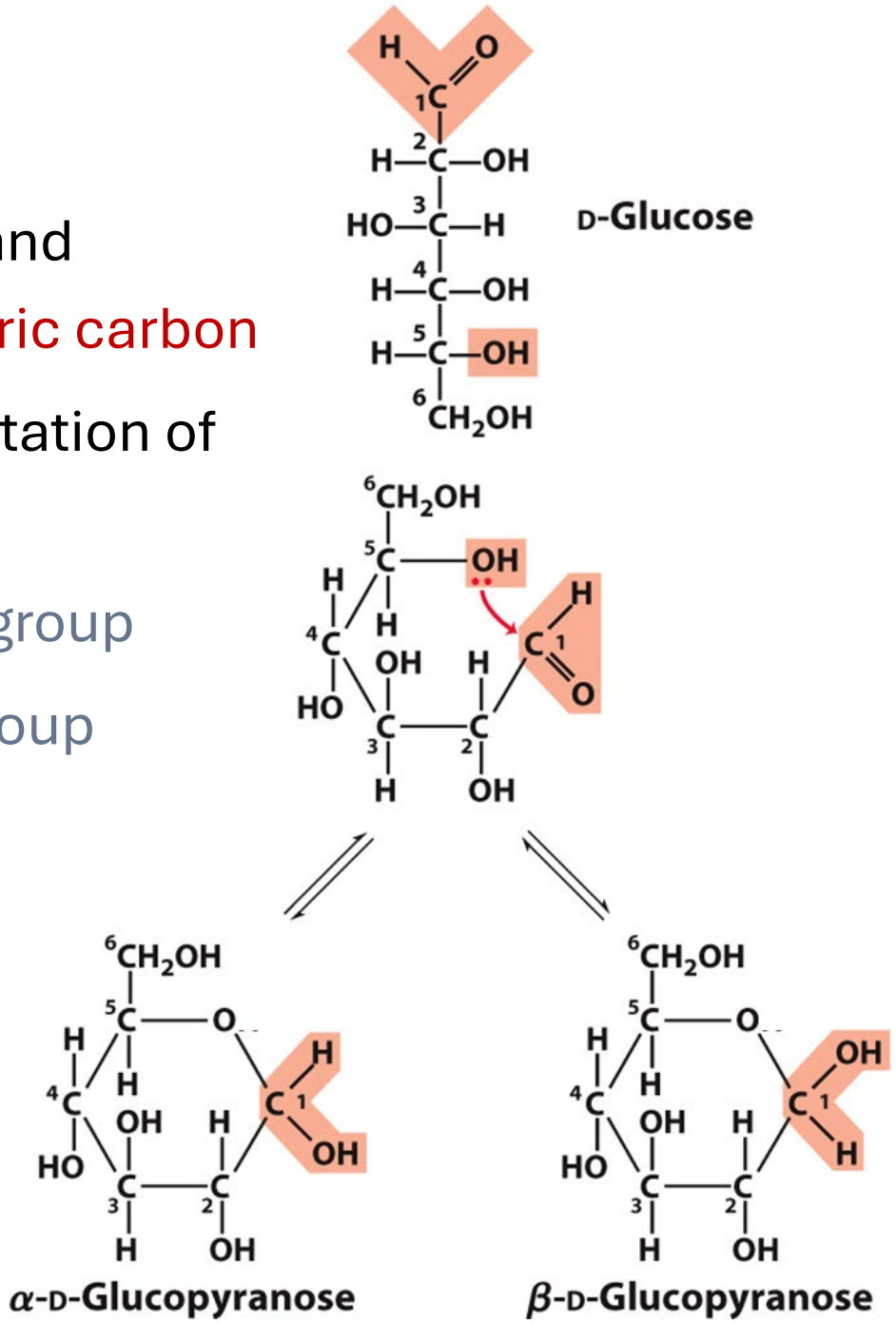


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Haworth projection of cyclic monosaccharides



Watch a video about Haworth projections

Conformation of cyclic monosaccharides

- Pyranose ring structures can occur in many different conformations

  chair, boat, envelope …

- Chair conformations are the most favorable ones

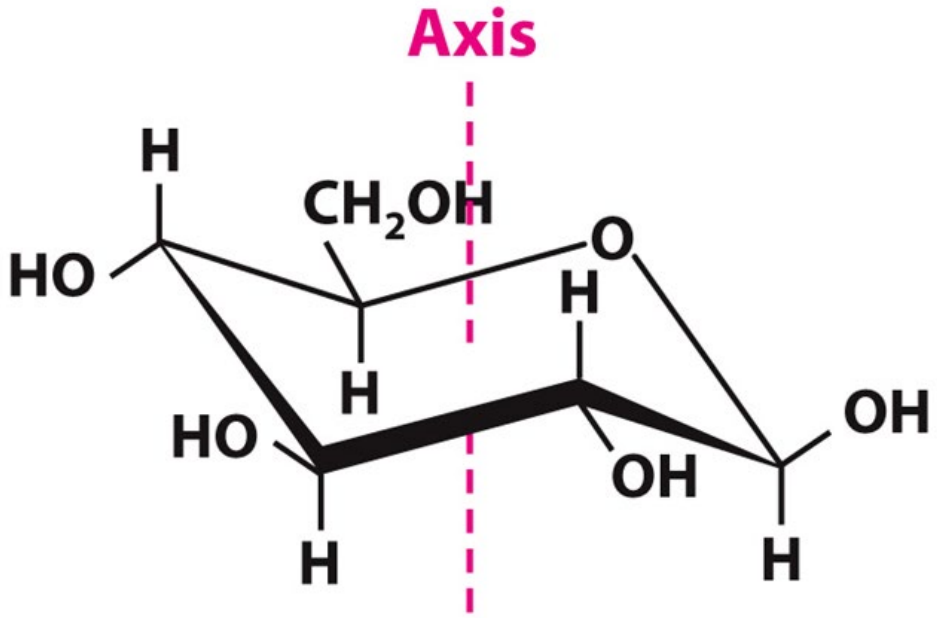


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Monosaccharide derivatives

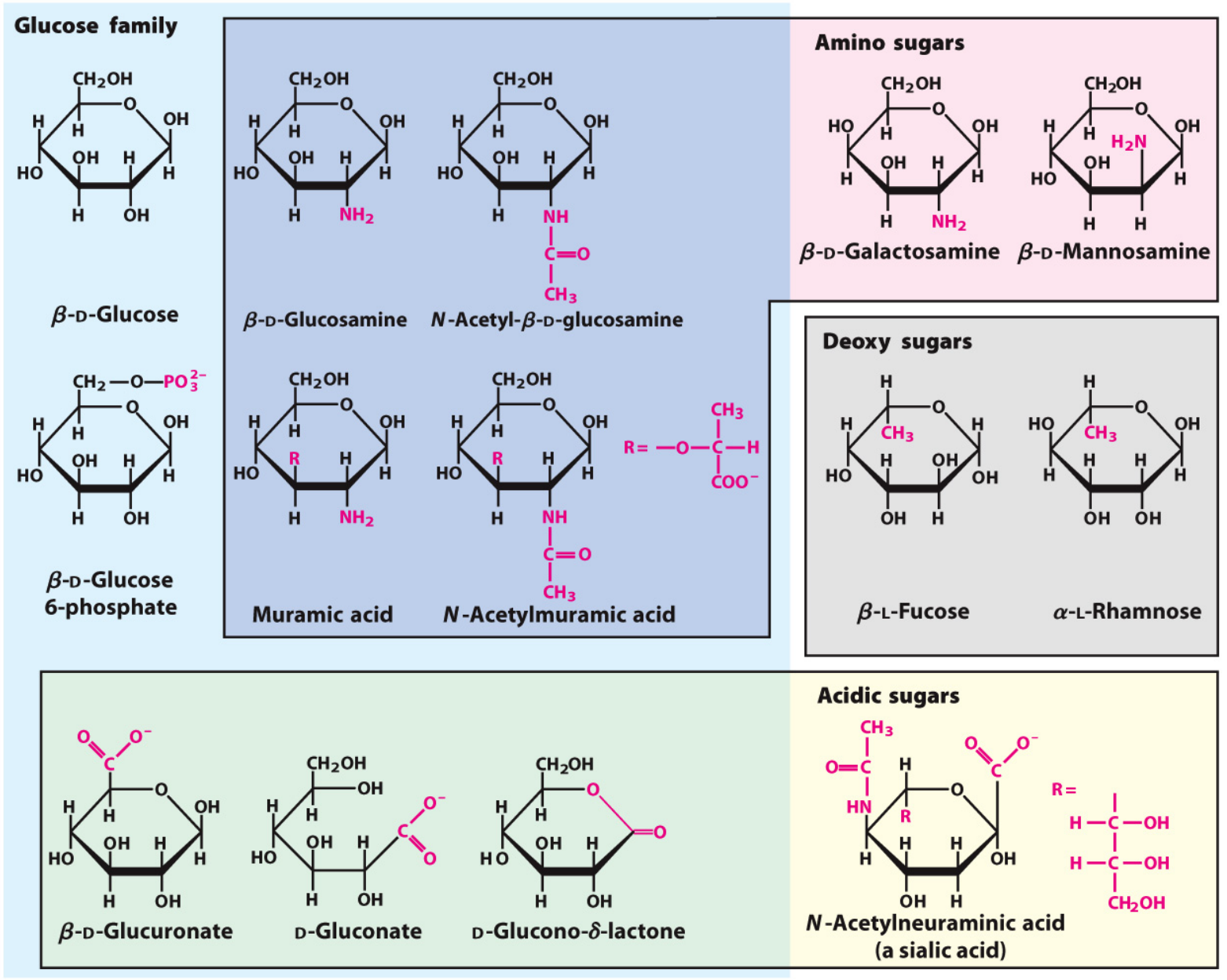
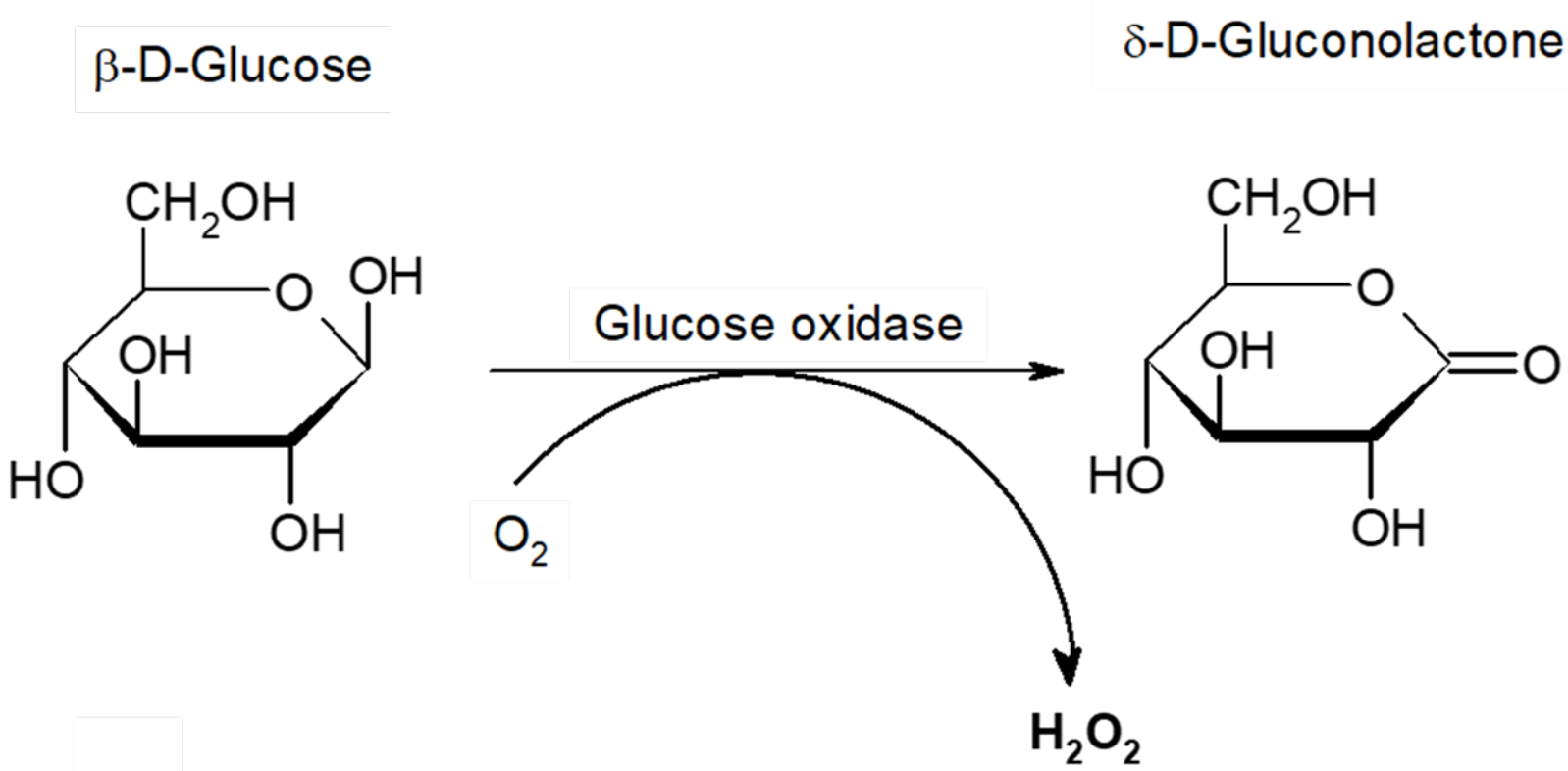


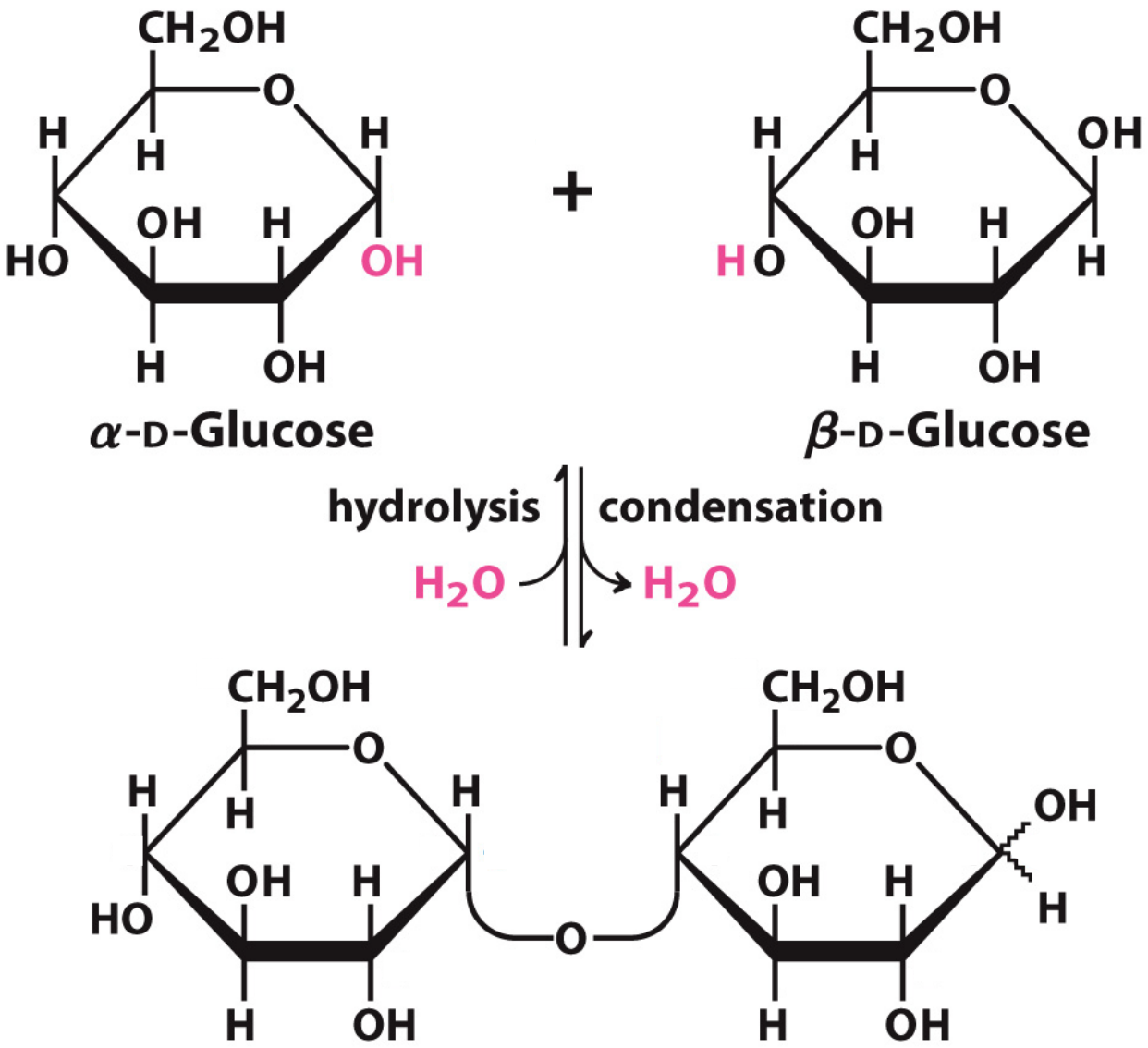
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Blood glucose test





Saccharide polymers



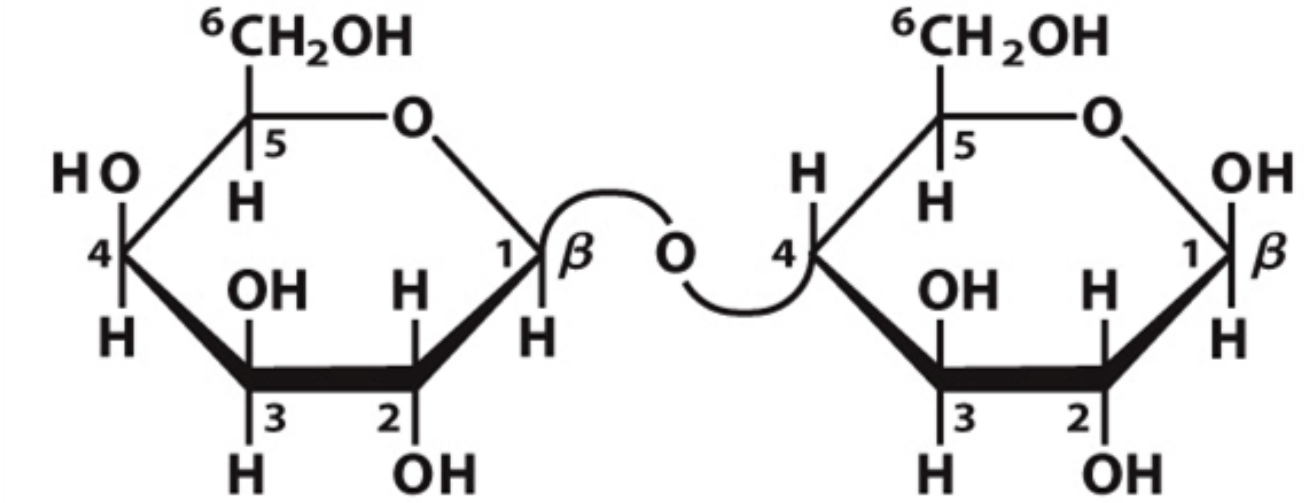
Two monosaccharides can be joined via an O-glycosidic bond between an anomeric carbon and a hydroxyl group

Figure 7-10
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Disaccharides

- A disaccharide is characterized by:
  - the two monosaccharides
  - the carbon atoms involved in the O-glycosidic bond
  - the anomeric configuration (α o β) of the monosaccharides
- Nomenclature:
  - Monosaccharide with the anomeric ending in *-osyl*
  - Configuration and carbon position the O-glycosidic bond
  - Second monosaccharide: ending in *-ose* if its anomeric carbon is free; or in *-oside* if it also participates in the O-glycosidic bond

Lactose

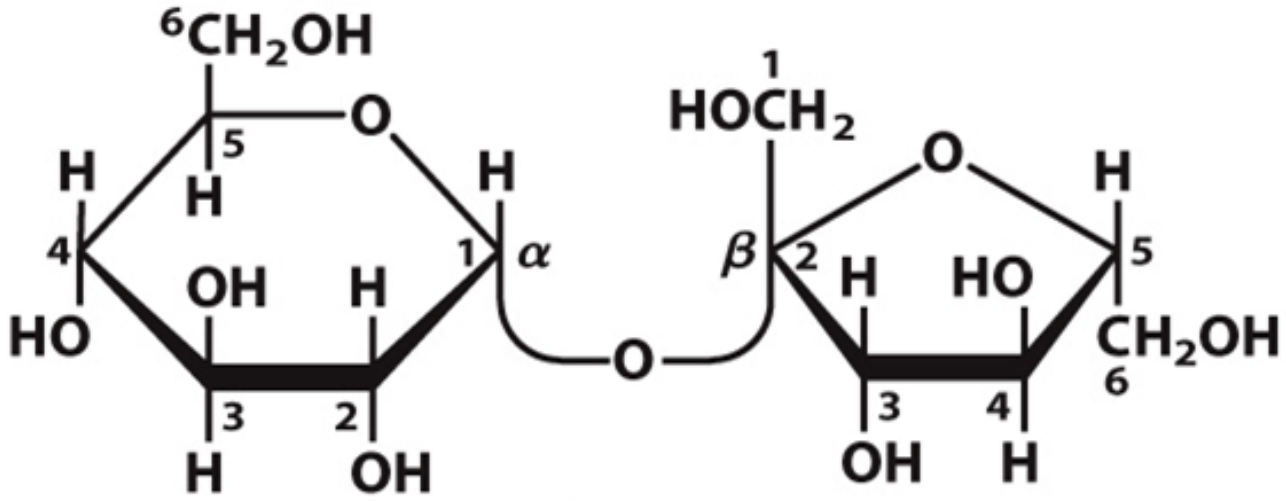


O-β-D-galactopyranosyl-(1→4)- β -D-glucopyranose

Gal(β1→4)Glc

Figure 7-11
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Sucrose



O-α-D-glucopyranosyl-(1→2)-β-D-fructofuranoside

Glc(α1→2)Fru

Figure 7-11
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Polysaccharides

- Polymers of many monosaccharides joined by O-glycosidic bonds

  the most common form in which carbohydrates are found

  they do not have a defined molecular weight

- Main functions:
  - Energy store
  - Structural support
- They can be classified as:
  - Homopolysaccharides or heteropolysaccharides
  - Linear or branched polymers

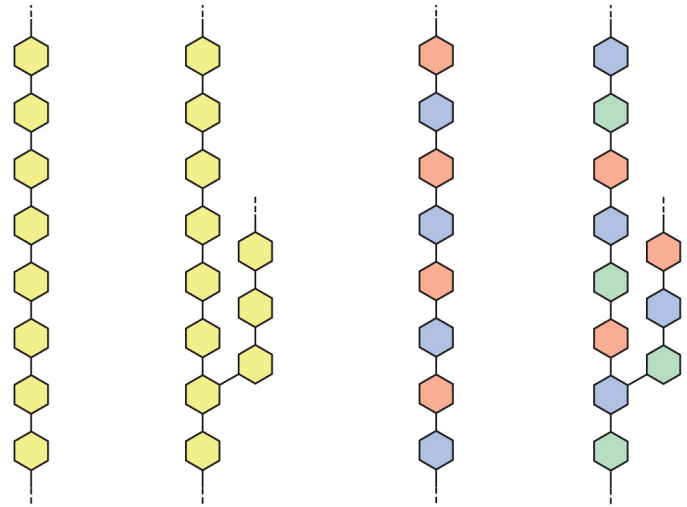
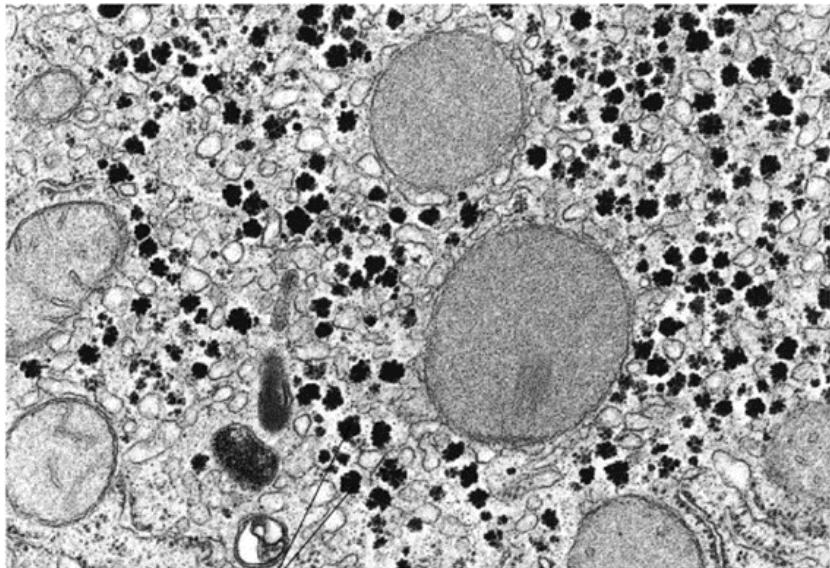


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Glycogen

- Branched homopolysaccharide of glucose residues
- Monomers are linked by $\alpha 1 \rightarrow 4$ bonds
- Branch points use $\alpha 1 \rightarrow 6$ linkages every 8-12 residues
- Molecular weight can reach several million datons
- Main storage polysaccharide in animals and many bacteria



Glycogen

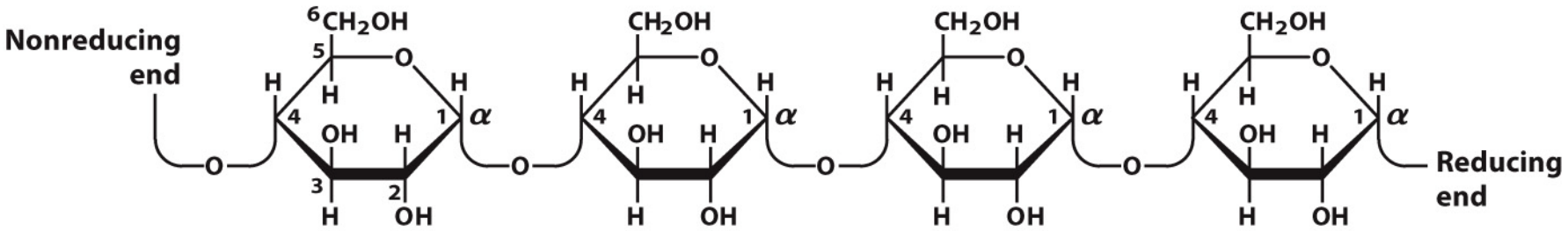


Figure 7-13a
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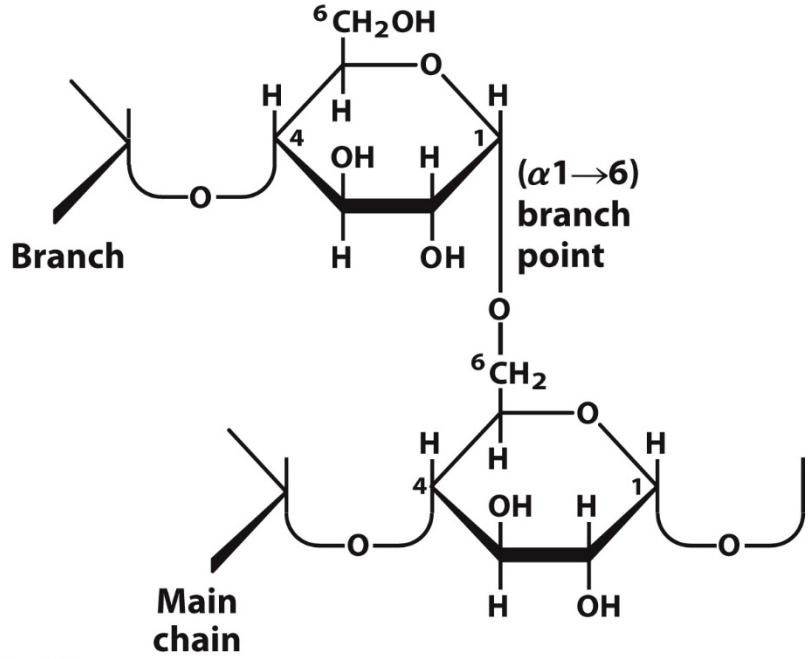


Figure 7-13b
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Glycogen

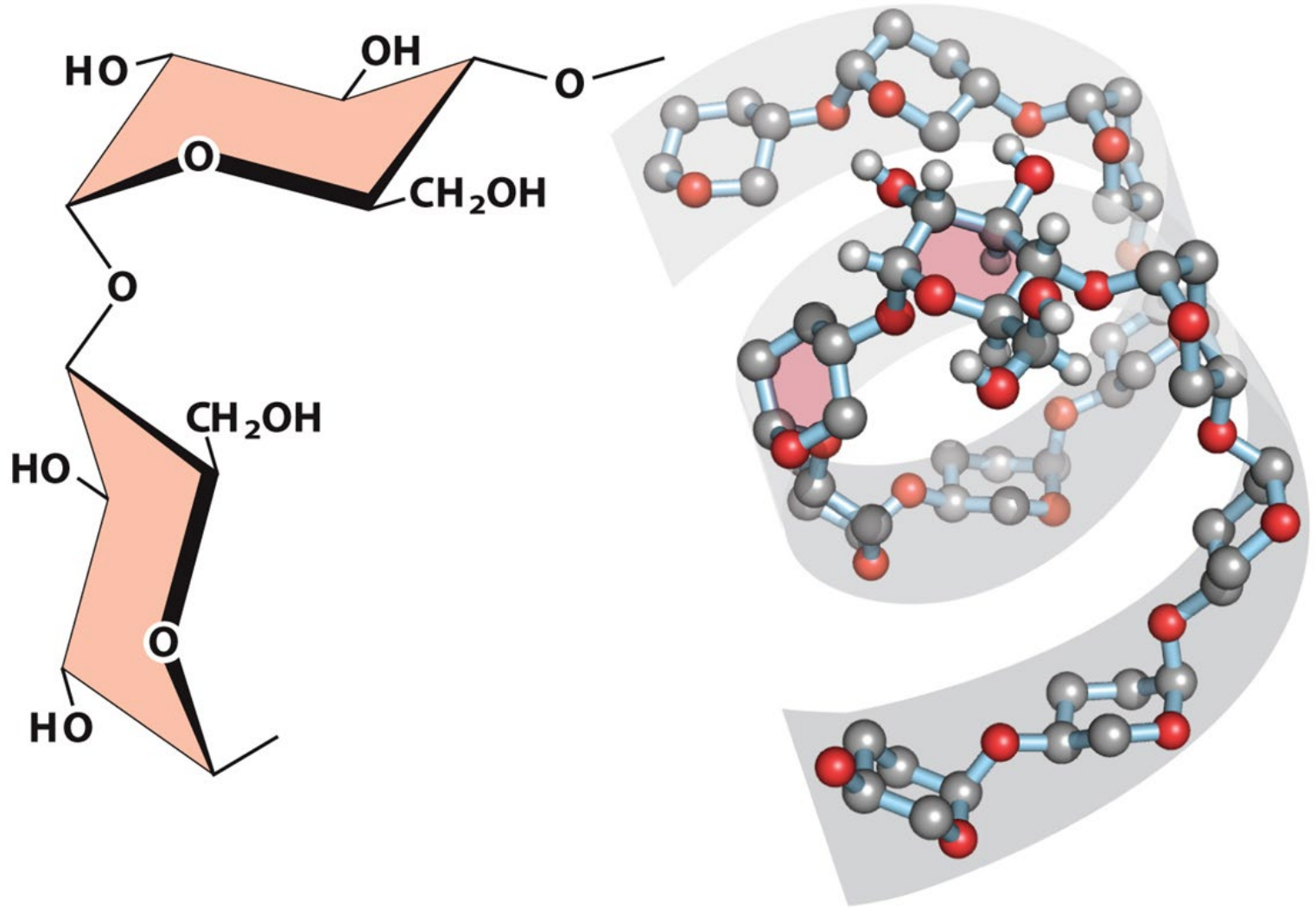
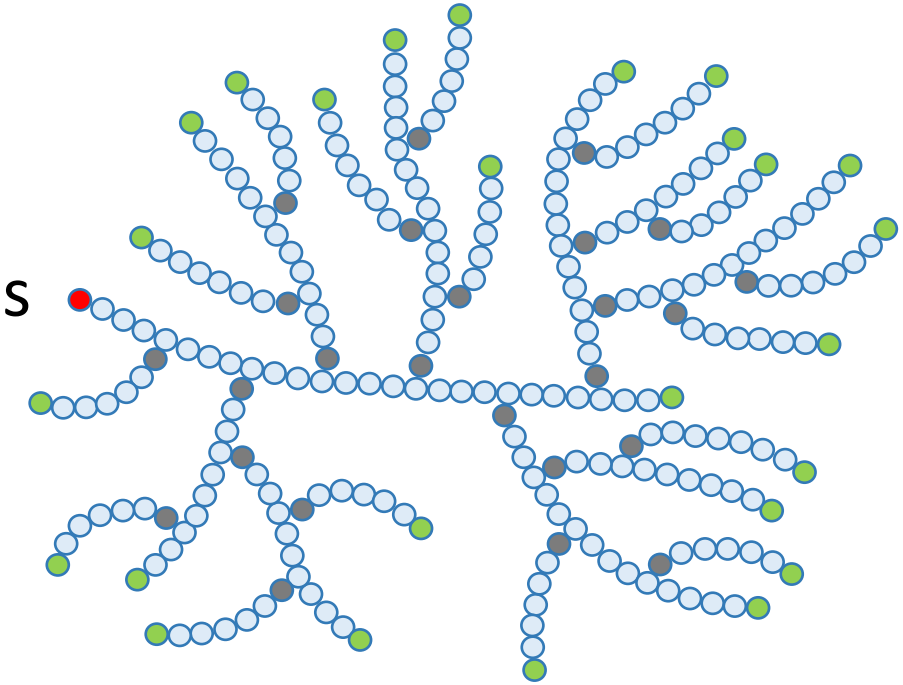


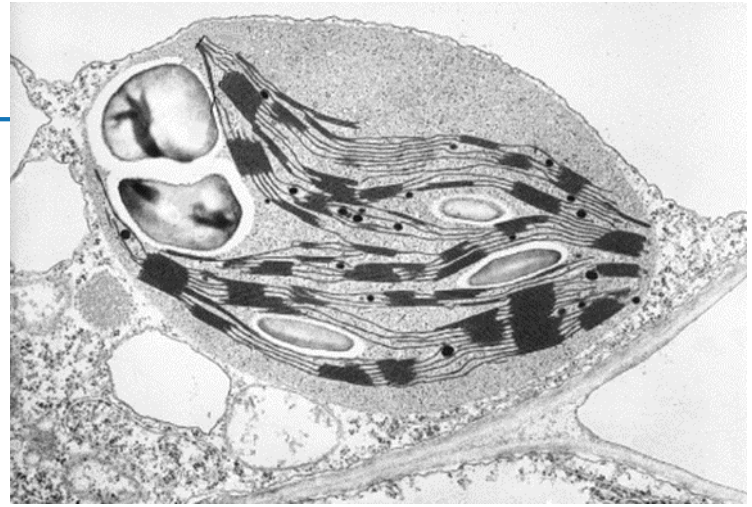
Figure 7-19
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Glycogen metabolism

- Glycogen is stored as cytoplasmic granules
- Granules contain enzymes that synthesize and degrade glycogen acting in the nonreducing ends
- A glycogen molecule has one reducing end but many non-reducing ends, so enzymatic processing occurs simultaneously at many sites



Starch



Main storage polysaccharide in plants

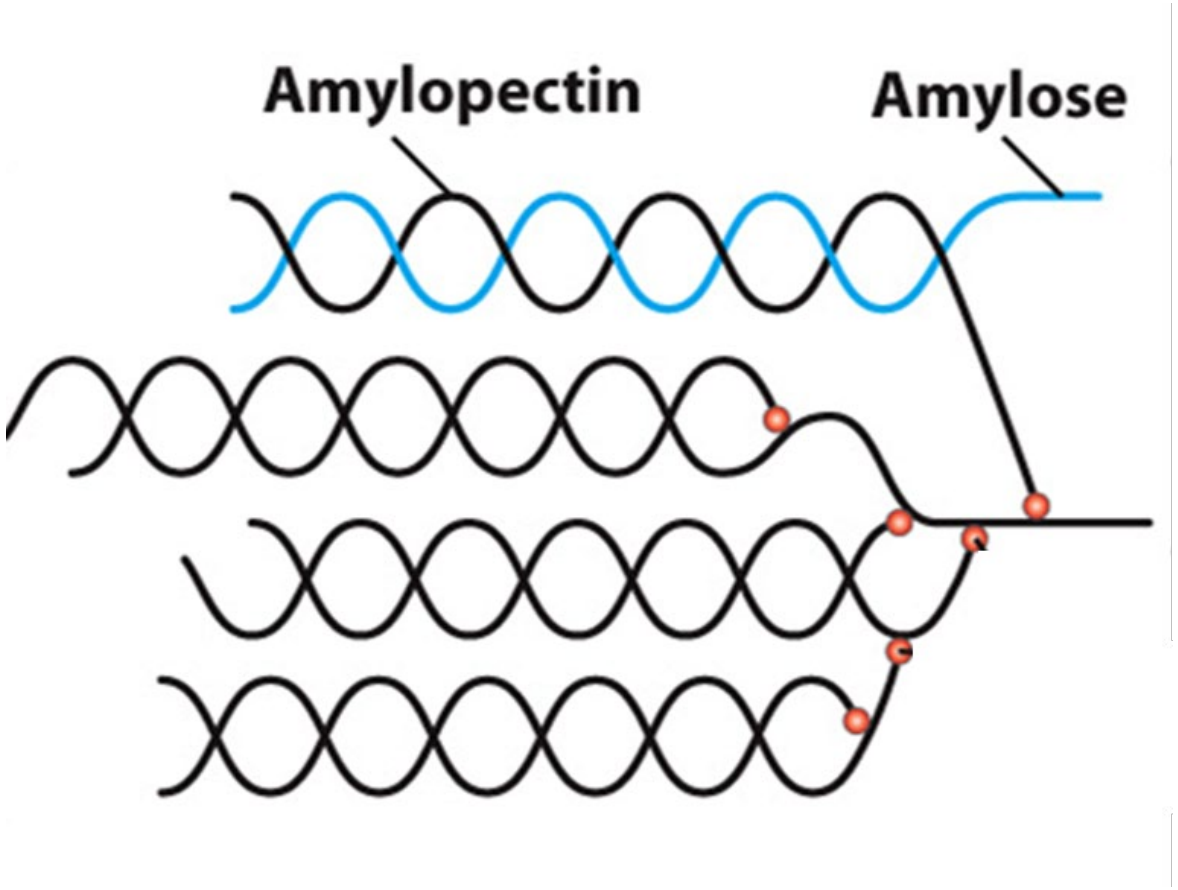


Figure 7-13
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Cellulose

- Most abundant organic polymer in nature
- Linear homopolysaccharide of glucose linked by β1 → 4 bonds
- Hydrogen bonds between neighboring residues and adjacent chains form insoluble fibers, providing structural support

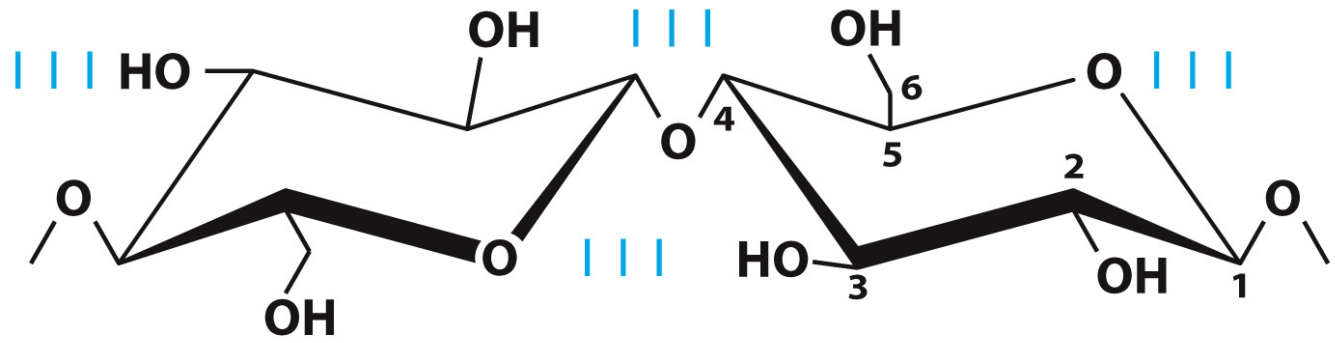


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Cellulose metabolism

- Cellulose is difficult to degrade because of its fibrous, water-insoluble structure
- Most animals cannot use cellulose as a fuel because they lack enzymes that catalyze the hydrolysis of $\beta 1 \rightarrow 4$ linkages
- Fungi, bacteria and protozoa secrete cellulase that allows them to use plant material as a glucose source
- Ruminants and termites live symbiotically with cellulase-producing microorganisms

Chitin



- Linear homopolysaccharide of N-acetylglucosamine linked by β1 → 4 bonds
- Forms extended fibers similar to cellulose
- Found in exoskeletons of insects, spiders, crabs, and other arthropods. Also found in fungal cell walls

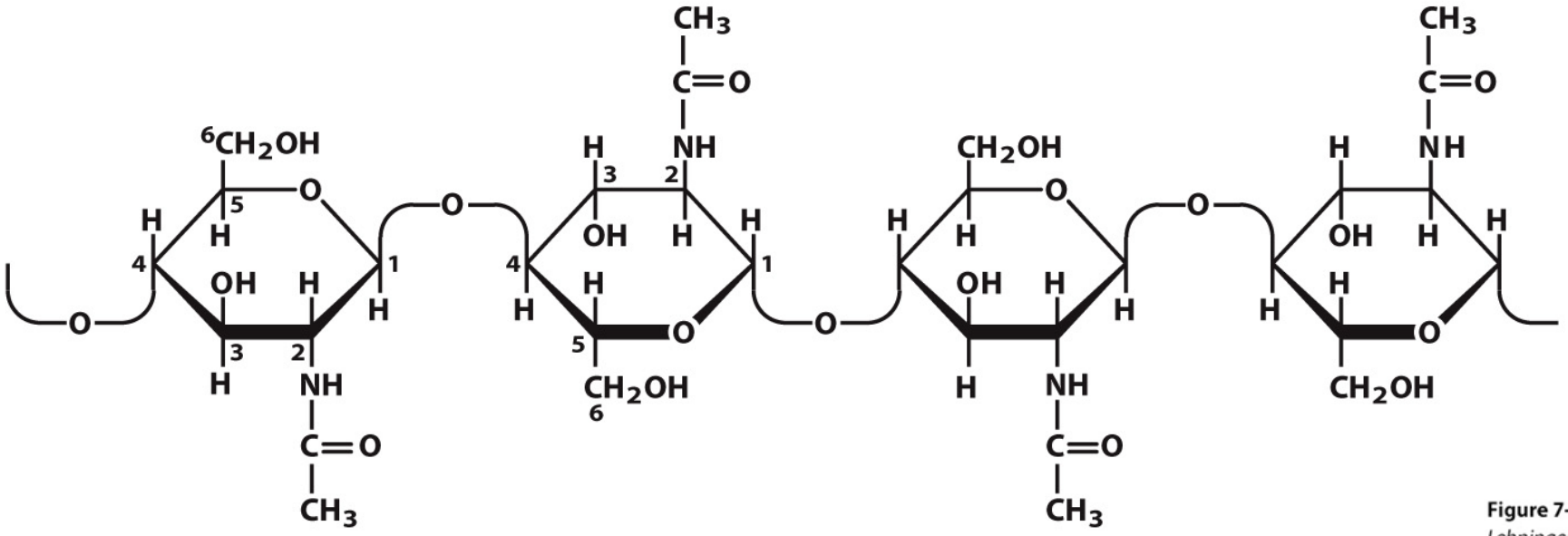
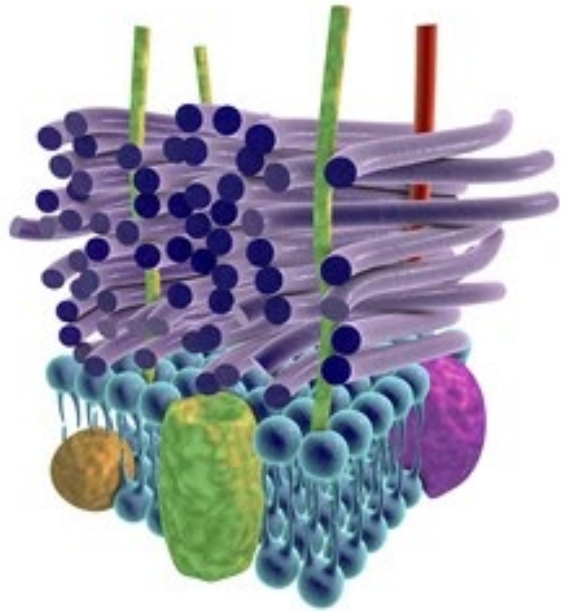


Figure 7-16a
Lehninger Principles of Biochemistry, Seventh Edition
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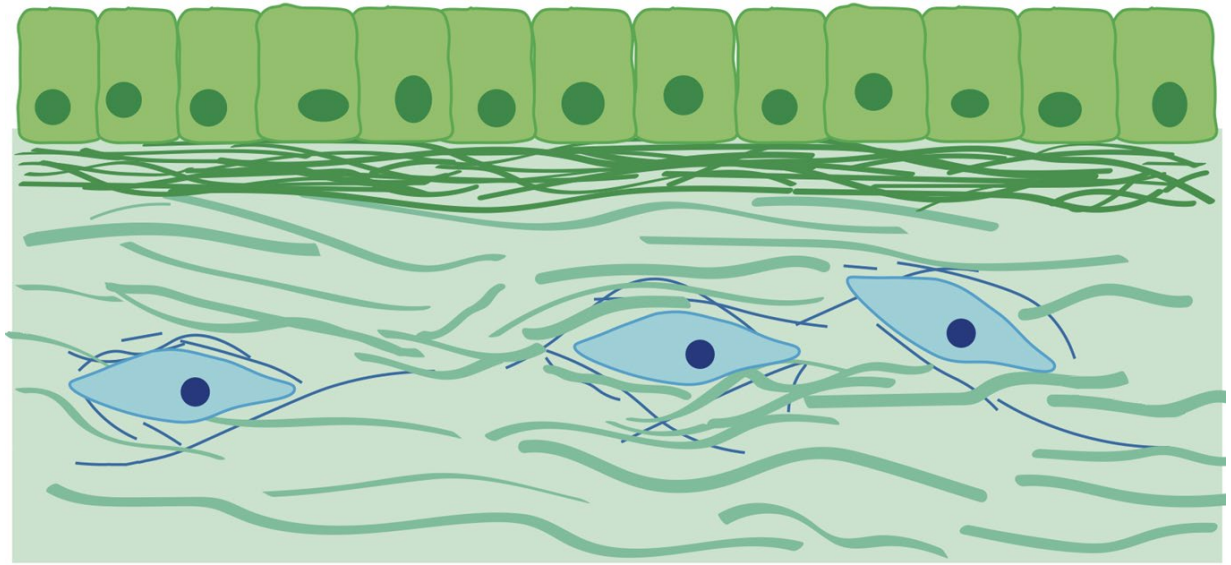
Peptidoglycan

- Rigid component of bacterial cell walls that prevents swelling and lysis
- Linear heteropolysaccharide of N-acetylglucosamine and N-acetylmuramic acid linked by β1 → 4 bonds. Linear polymers are cross-linked by short peptides
- The enzyme lysozyme kills bacteria by hydrolyzing the glycosidic bond between GlcNAc and MurNAc



Carbohydrates in the extracellular matrix

- Together with fibrous proteins, carbohydrates provide strength, elasticity, and a physical barrier in tissues
- Carbohydrates in the extracellular matrix include:
  - Glycosaminoglycans
  - Glycoconjugates
    - proteoglycans
    - glycoproteins
    - glycolipids



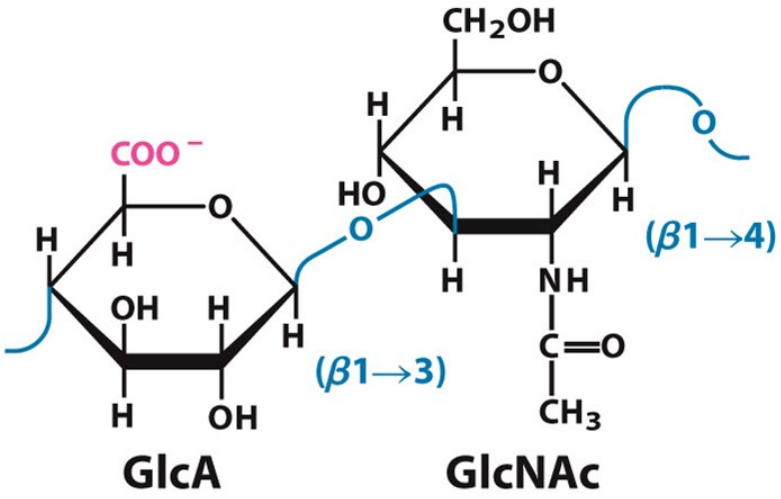
Glycosaminoglycans

- Linear polymers of repeating disaccharide units

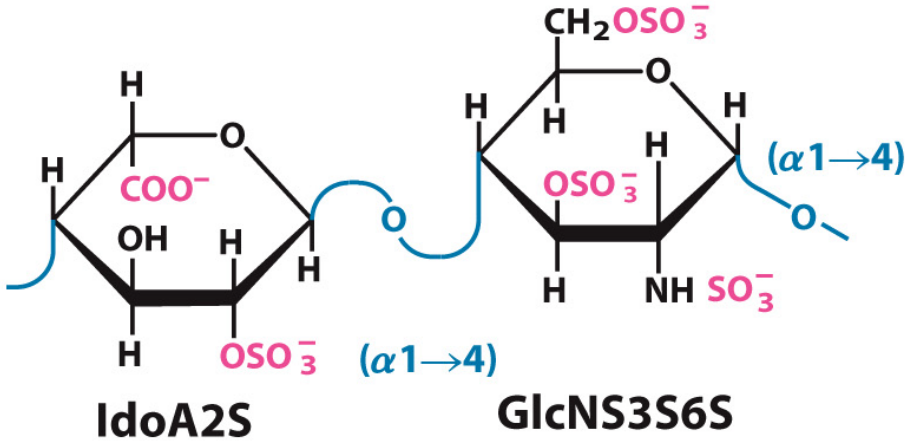
  most contain a uronic acid and an amino sugar

- Their negative charges attract water and counterions, generating hydrated gels

  they increase viscosity and provide elastic support to the extracellular matrix

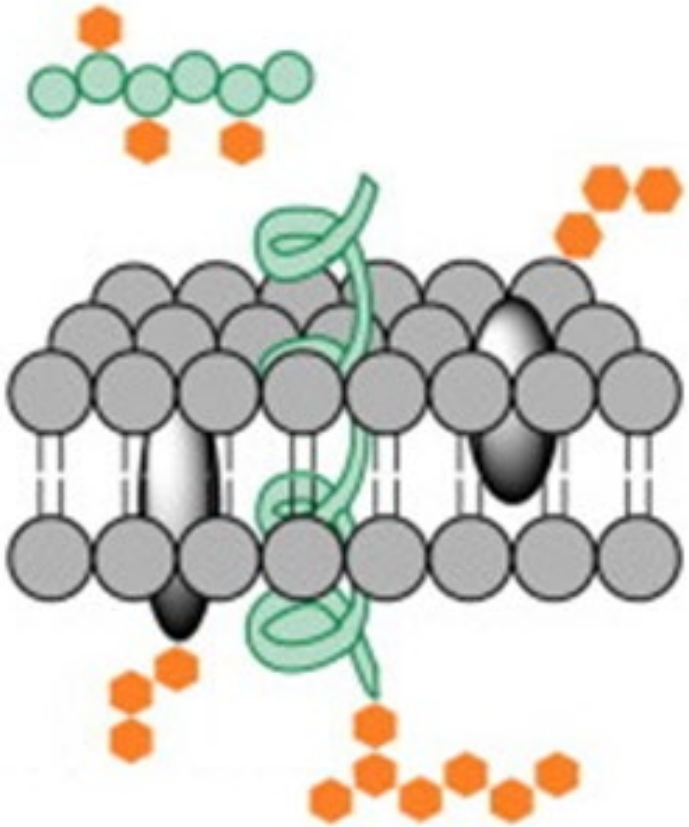


Hyaluronate



Heparin

Glycoconjugates



- Proteoglycans
- Glycoproteins
- Glycolipids

We will study them in Unit III